

Mathematical Methods for International Commerce

Week 12/1: Indefinite Integration (6.1)

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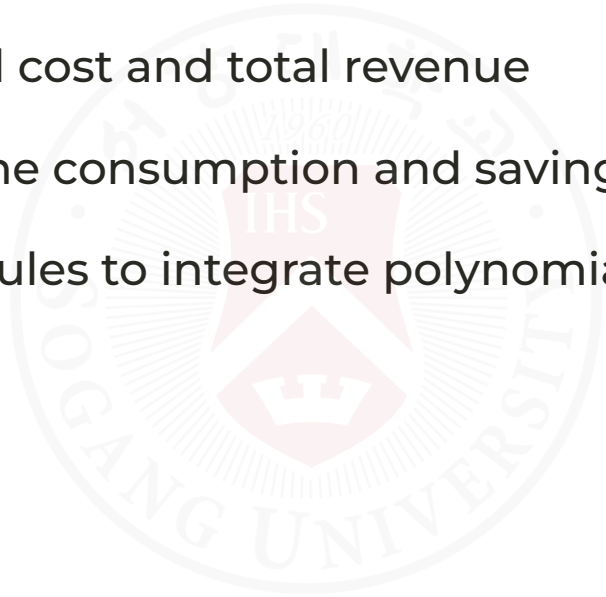
May 19, 2026

Why It Matters in Economics & Finance

- Integration helps in determining total cost, total revenue, and accumulated savings over time.
- It is crucial in calculating areas under curves, which represent total accumulated quantities.
- Applications include:
 - Finding total cost from marginal cost
 - Determining total revenue from marginal revenue
 - Estimating accumulated savings from marginal propensity to save

Learning Objectives

- Recognize the notation for indefinite integration
- Integrate power functions and exponential functions
- Apply integration to find total cost and total revenue
- Apply integration to determine consumption and savings functions
- Use inspection and linearity rules to integrate polynomial and exponential functions



Understanding Indefinite Integration

- Indefinite integration is the reverse of differentiation.
- The general form:

$$\int f(x) dx = F(x) + C$$

where:

- $f(x)$ is the integrand
- $F(x)$ is the antiderivative
- C is the constant of integration

Note: The antiderivative, also known as the indefinite integral, is a function whose derivative equals the original function $f(x)$. In essence, finding the antiderivative reverses the process of differentiation.

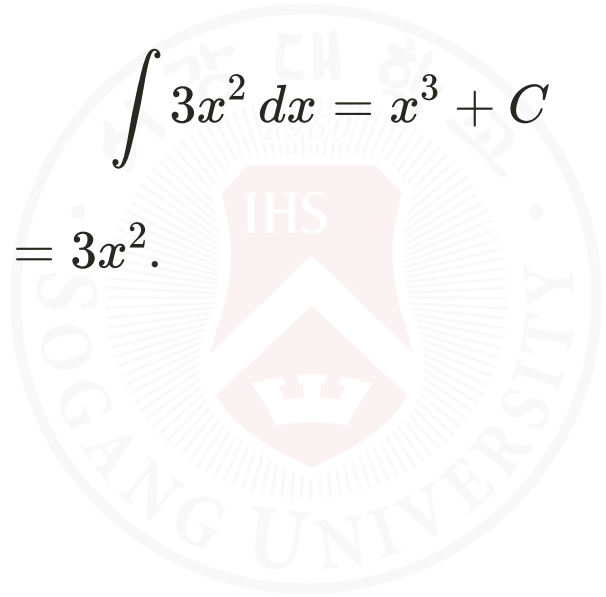
Understanding Indefinite Integration (cont.)

- Example:

- If $f(x) = 3x^2$, then:

$$\int 3x^2 dx = x^3 + C$$

- Why? Because $\frac{d}{dx}(x^3 + C) = 3x^2$.



The Constant of Integration

- Why do we add $+C$?
 - Represents an **arbitrary constant** since differentiation eliminates constants.
- Example:
 - $\int 3x^2 dx = x^3 + C$
 - Both $x^3 + 5$ and $x^3 - 7$ differentiate to $3x^2$.
- From economics perspective, C often represents fixed costs or baseline values in economic applications.
- In practice, we often determine C using initial conditions or boundary values.

Indefinite Integration: Water Tank Analogy

1. Water Flow = Derivative (Rate of Change)

- Water flows into the tank at a changing speed.

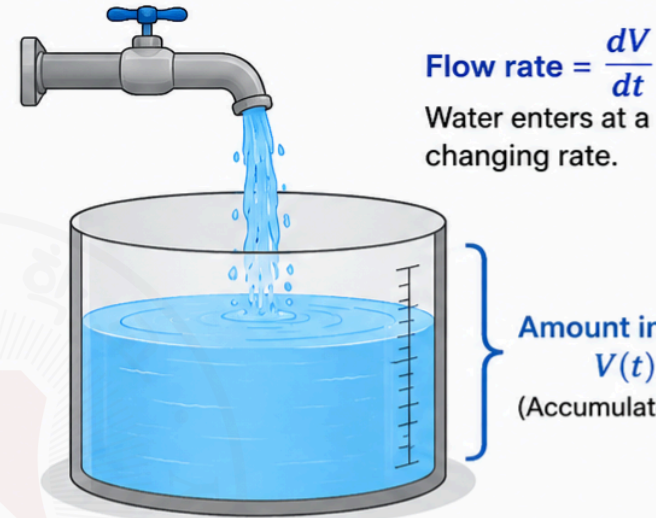
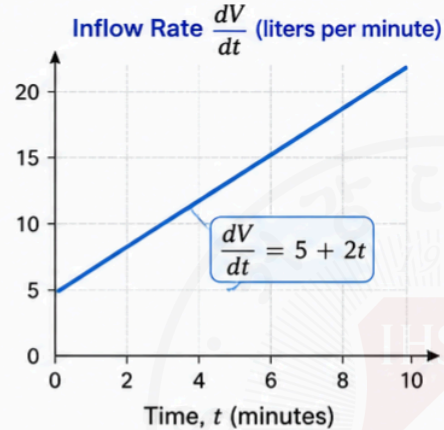
- Flow rate: $\frac{dV}{dt}$

This tells us how fast water enters (the instantaneous change).

- Example:

$$\frac{dV}{dt} = 5 + 2t$$

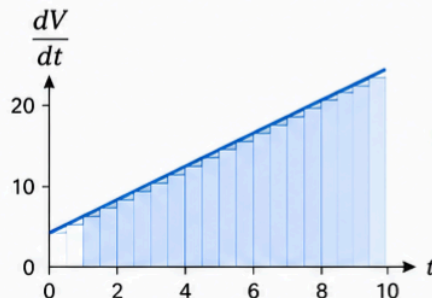
At higher t , water enters faster.



Amount in tank
 $V(t)$
(Accumulated water)

2. Accumulation = Integration

- The total amount of water in the tank is found by adding all small inflows together.



$$V(t) = \int (5 + 2t) dt$$
$$= 5t + t^2 + C$$

Where:

$V(t)$ = total accumulated water
 C = initial amount already in the tank

Area under the flow rate curve = total water accumulated

3. Economic Interpretation

The same idea appears everywhere in economics and finance:

Marginal / Rate		Total Accumulation
 Marginal Revenue (MR)	→	Total Revenue (TR)
 Marginal Cost (MC)	→	Total Cost (TC)
 Investment Flow	→	Total Investment
 Inventory Added per Day	→	Total Inventory

Key Idea:

Derivatives measure the rate of change.
Integrals measure accumulated totals.

Power Rule for Indefinite Integrals

- General formula:

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$

- Examples:

- $\int x^5 dx = \frac{x^6}{6} + C$

- $\int x^{-3} dx = \frac{x^{-2}}{-2} + C = -\frac{1}{2x^2} + C$

Integration of Exponential Functions

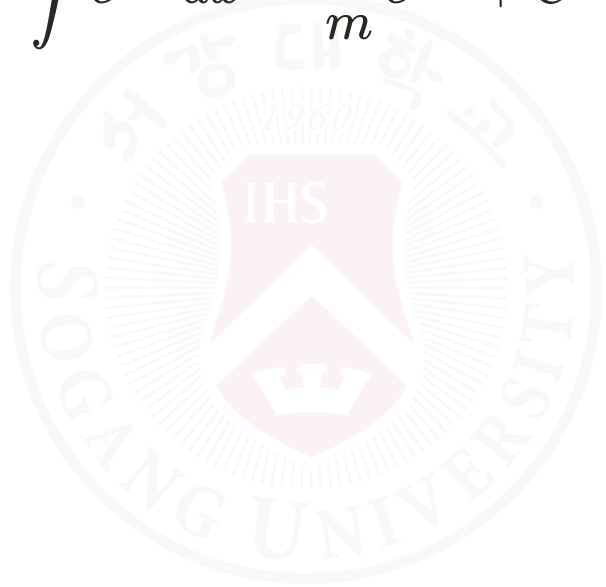
- Formula:

$$\int e^{mx} dx = \frac{1}{m} e^{mx} + C$$

- Example:

- $\int e^{2x} dx = \frac{1}{2} e^{2x} + C$

- $\int e^{-0.5x} dx = -2e^{-0.5x} + C$



Practical Example: Marginal Revenue to Total Revenue (1)

- Suppose the **marginal revenue function** is given by:

$$MR(x) = 50 - 2x$$

- To find the total revenue function:

$$\begin{aligned} TR(x) &= \int (50 - 2x) dx \\ &= 50x - x^2 + C \end{aligned}$$

- **Economic Interpretation:**
 - The area under the MR curve represents the total revenue.

Practical Example: Marginal Revenue to Total Revenue (2)

- If the marginal revenue is given by:

$$MR(x) = 12 - 2x$$

- The total revenue function is:

$$TR(x) = \int (12 - 2x) dx = 12x - x^2 + C$$

- Interpretation:
 - The total revenue function represents the total income generated from selling x units of a good.
 - The constant C represents the initial value of total revenue when output is zero.
 - The area under the marginal revenue curve gives the total revenue.
 - The total revenue function changes quadratically with respect to x .

Practical Example: Marginal Cost to Total Cost (1)

- If the marginal cost is given by:

$$MC(x) = 3x^2 + 5$$

- The total cost function is found by integrating:

$$TC(x) = \int (3x^2 + 5) dx = x^3 + 5x + C$$

Practical Example: Marginal Cost to Total Cost (2)

- Given the marginal cost function:

$$MC(x) = 4x^2 - 3x + 7$$

- Integrate to find the total cost function:

$$TC(x) = \int (4x^2 - 3x + 7) dx = \frac{4x^3}{3} - \frac{3x^2}{2} + 7x + C$$

Economic Example: Consumption & Savings

- Given the marginal propensity to consume (MPC):

$$MPC = 0.8$$

- Integrate to find the consumption function:

$$\int 0.8 dY = 0.8Y + C$$

- If $MPS = s$, then integrating gives the savings function $S(Y) = sY + C$.

Method of Inspection

- For more complex integrals, identify a form that simplifies to a known pattern:

$$\begin{aligned}\int (5x^2 + 3x + 2) dx &= \int 5x^2 dx + \int 3x dx + \int 2 dx \\ &= \frac{5x^3}{3} + \frac{3x^2}{2} + 2x + C\end{aligned}$$

Practice Problems

1. Integrate:

- (a) $\int 4x^3 dx$
- (b) $\int e^{2x} dx$

2. Find the total cost given $MC = 5x^2 - 3x + 4$.

3. Determine the total revenue function given $MR = 15 - 0.5x$.

4. Given the marginal propensity to consume is $MPC = 0.8$, determine the consumption function if initial consumption is 100.

Summary

- Indefinite integration helps in finding total values given marginal functions.
- Applications in economics include calculating total cost, total revenue, and accumulated savings.
- Key formulas:
 - Power Rule
 - Exponential Function
 - Method of Inspection



The logo of Gangneung University is a circular seal. It features a central shield with a crown on top and the letters 'IHS' inside. The year '1960' is inscribed above the shield. The outer ring of the seal contains the university's name in Korean and English.

2. Group Activity

Math Battle: Indefinite Integrals

Objective: Apply indefinite integration concepts to solve economic problems.

Format: 6 groups, 4-5 students each. Each group receives a distinct problem.

Time: 20 minutes to solve, 5 minutes to present.

Math Battle Rules:

- Each group presents their solutions.
- Other groups can challenge the solution if they find discrepancies.
- Correct challenges earn 2 points; incorrect challenges lose 1 point.
- Winning group gets a prize!

Group 1: Marginal Cost and Total Cost

A firm's marginal cost function is:

$$MC(x) = 12x^2 - 6x + 4$$

1. Find the total cost function by integration.
2. If fixed cost is \$80, determine the constant of integration.
3. Interpret the meaning of the fixed cost economically.

Group 2: Marginal Revenue and Total Revenue

A firm's marginal revenue function is:

$$MR(x) = 30 - 4x$$

1. Find the total revenue function by integration.
2. Assume total revenue is zero when output is zero. Determine the constant of integration.
3. At what output level does marginal revenue become zero?

Group 3: Consumption Function

The marginal propensity to consume is constant:

$$MPC = 0.75$$

1. Integrate to find the consumption function.
2. If autonomous consumption is \$120, determine the full consumption function.
3. Explain economically what the MPC means.

Group 4: Savings Function

The marginal propensity to save is:

$$MPS(Y) = 0.2$$

1. Integrate to find the savings function.
2. If initial savings are (-\\$50) when income is zero, determine the full savings function.
3. Interpret the meaning of negative initial savings and positive marginal propensity to save.

Group 5: Investment Accumulation

A firm's marginal investment function is:

$$MI(t) = 100e^{0.04t}$$

1. Integrate to find the total investment function.
2. Assume accumulated investment is zero at ($t = 0$). Determine the constant of integration.
3. Calculate the accumulated investment after 5 years.

Group 6: Marginal Benefit and Total Benefit

A consumer's marginal benefit function is:

$$MB(Q) = 20 - 2Q$$

1. Integrate to find the total benefit function.
2. Assume total benefit is zero when quantity is zero. Determine the constant of integration.
3. Explain why total benefit increases at a decreasing rate.

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3. Home work #2

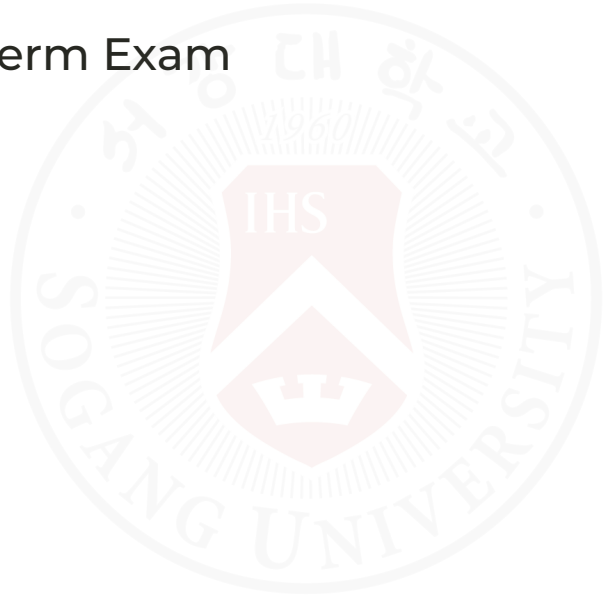
Homework #2

- **Due Date:** June 11, 2026, end of the day.
- **Submission Format:** Submit your solutions as a single PDF file via the Cyber Campus.
- **Instructions:**
 - Clearly show all steps and calculations.
 - Include explanations for your answers where applicable.
 - Ensure your submission is neat and well-organized.
 - Bring any questions to the office hours or email me.
- Work on your Home Assignment #2 (Jacques, 10th edition, Chapters 5-9):
 - Chapter 5.1: Exercise 5.1, Problem 5 (p. 411)
 - Chapter 5.2: Exercise 5.2, Problem 7 (p. 427)
 - Chapter 5.3: Exercise 5.3, Practice Problem inside the chapter (p. 432 only)
 - Chapter 5.4: Exercise 5.4, Problem 5 (p. 456)
 - Chapter 5.5: Exercise 5.5, Problems 6 and 7 (p. 467)
 - Chapter 5.6: Exercise 5.6, Problem 4 (p. 479)
 - Chapter 6.1: Exercise 6.1, Problem 3 (p. 506)
 - Chapter 6.2: Exercise 6.2, Problem 6 (p. 521)
 - Chapter 7.1: Exercise 7.1, Problem 5 (p. 553)
 - Chapter 7.2: Exercise 7.2, Problem 6 (p. 572)
 - Chapter 7.3: Exercise 7.3, Problem 5 (p. 584)
 - Chapter 8.1: Exercise 8.1, Problem 6 (p. 615)
 - Chapter 8.2: Exercise 8.2, Problem 3 (p. 626)
 - Chapter 9.1: Exercise 9.1, Problem 4 (p. 655)
 - Chapter 9.2: Exercise 9.2, Problem 3 (p. 670)

Good luck!

Next Classes

- (May 21) Definite Integration (6.2)
- (May 26) Quiz #2
 - Covers materials after Midterm Exam





Any QUESTIONS?

Thank you for your attention and participation!